

The Expanded Brain

Medicine uses electrode implants, magnetic fields, radio waves, and chemicals to treat brain problems. These technologies could also potentially enhance normal brain functions.

Boosting the brain

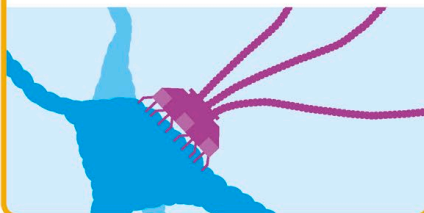
“Overclocking” is the speeding up of a computer’s internal clock, which coordinates all its circuits, to push components to work faster and harder. Like computers, the brain uses tiny electrical signals in the form of nerve impulses, which raises the possibility that it might be similarly overclocked. Depending on the region stimulated, this might improve attention and focus, information processing, and memory.

IS IT SAFE TO SPEED UP YOUR BRAIN?

So far, the evidence suggests that tDCS is safe. Thousands of healthy people have taken part in experiments using tDCS, and no adverse effects have been noted.

NANO NEUROBOTS

Researchers are developing almost molecule-sized, robotlike implants, to deliver medical drugs, for example. Next-generation neurobots that are specialized to deliver programmed electronic signals might also accelerate both the way neurons work and how they process their nerve impulses.



A HIPPOCAMPAL PROSTHESIS CAN IMPROVE MEMORY PERFORMANCE BY AS MUCH AS 37 PERCENT

Transcranial direct current stimulation

In transcranial direct current stimulation (tDCS), a direct electric current is passed at a constant low strength through the brain, between padlike electrodes attached to the skin. Sessions of tDCS have helped treat depression and relieve pain. The ability of tDCS to enhance a range of cognitive functions, from creativity to logical reasoning, is being researched. Here, tDCS is shown in use at the same time as TMS, although the techniques are not actually used simultaneously.

Inhibiting the brain

During cathodal tDCS, the current is negative with respect to the brain’s own electrical activity. This has the effect of slowing or inhibiting nerve cells, for example, to reduce hyperactivity.

